

Jon Edwards

Principal Software Engineer/Engineering Manager

Product Engineering · Data Platforms · AI · Scaling Systems · Team Leadership

(360) 509-8185 • jkedwards@me.com • US Person • Bainbridge Island, WA

Summary

Principal Software Engineer building streaming data platforms over datasets in the hundreds of millions of records, modernizing developer platforms with Nix and monorepo tooling, and operationalizing AI-assisted engineering practice. In the past year, shipped a Flow-Based Programming data graph (~2.5M events/day and growing), avmbench — a property and people delivery platform that unlocked \$1M in net new revenue and new categories of products — and Scala load shedding for a high-traffic search layer. 10+ years bridging deep IC and engineering leadership; has managed teams of up to 12.

Skills

- **Specialties:** streaming data platforms, developer-platform modernization, AI-assisted engineering practice, Flow-Based Programming, distributed data pipelines
- **Leadership:** team management, mentorship, high-volume substantive code review, engineering governance
- **Languages:** Python, TypeScript, JavaScript, CUE, Bash; working knowledge of Rust, Scala 2, Java, C#, Kotlin, Swift
- **Cloud & Infrastructure:** AWS, Kubernetes, Helm, Terraform, Azure
- **Build & Deploy:** Nix, Docker, monorepos, CI/CD pipelines (GitHub Actions, Jenkins)
- **Data & AI:** Databricks, Delta Lake, Parquet, PostgreSQL, Redis, Elasticsearch, MCP, Claude API
- **Frontend & Mobile:** React, Vue, React Native, Angular 2+, Cordova

Work Experience

Whitepages

(consumer search platform: 1B+ indexed pages, 10M+ unique users/month)

Principal Software Engineer — Jun 2023 – Present

- Architected and productionized a data graph streaming platform grounded in J. Paul Morrison's Flow-Based Programming, replacing parts of a multi-week, manually coordinated release process with on-demand delivery; currently handles ~2.5M events/day across two tenants in active onboarding.

- Backing datasets cover ~355M people, ~436M phones, ~483M emails, and ~234M addresses; the platform also powers a real-time pipeline streaming scraped public-records data to subscriber webhooks.
- Own architecture decisions, the function server, the pipeline compiler (built on a CUE-based DSL), metrics and logging, and bulk ingestion/export operations.
- Built avmbench, a Property and People data delivery platform that unlocked a \$1M net new revenue source — covering ~125M properties across all 50 states with a 17.1% national median error rate on valuation estimates, built on a custom geographic aggregation DSL and Census TIGER pipelines.
- Designed and shipped P95-latency-based load shedding for the search layer in Scala — ring-buffer windowed computation, OPEN/TRIPPED state machine with hysteresis, 429-with-Retry-After rejection — wrapping all 18 route handlers.
- Led migration of 2.5M events/day from Amplitude to Databricks Delta tables, eliminating several thousand dollars per month in overage fees while preserving the Salesforce/Kinesis ingest path.
- Drove monorepo migration with shared tooling, standardized linting, Nix-based developer environments (wp-home-manager), and an internal developer-productivity CLI (pagecraft, PR review workflow with parallelized GitHub fetches and Nix/darwin tooling) — supporting ~100 service releases per week and shifting CVE scanning from infrequent reactive SRE scans to proactive CI gates.
- Built MCP integration and AI inference harness on the engineering platform; authored org-wide documentation to set AI-assisted engineering practice.
- Combined hands-on mentorship and operationalized agentic AI with a robust spec process to lift delivered work 2.5–4x year-over-year — CRO tickets, bug fixes, feature requests, and new tooling — leaving the team running ahead of business demand with real operational headroom, despite shrinking headcount.
- Sustained high-volume, substantive peer code review across the platform org — 261 reviews in the first 15 weeks of 2026 alone (~75/month, mean 6 / median 4 comments per PR, ~1,000-line PR average) — reinforcing engineering standards and a small-PR discipline.
- *Technology used:* Python, Rust, Scala, CUE, Nix, Kubernetes, Helm, Databricks, Delta Lake, MCP.

Lead Sr. Software Engineer, Engineering Manager — Dec 2021 – Jun 2023

- Reversed a multi-quarter decline in Google indexation, increasing indexed page count by 30%.
- Drove dozens of millions of SERP-indexed URLs from red LCP (Core Web Vitals failure) to zero, with monitoring automation to keep them green.
- Increased mobile revenue by 20% YoY by focusing on stability and adding new features.
- Lifted mobile app release cadence from every other week to several times per week, focused on SEO performance and mobile feature delivery.

- Managed and mentored a team of 6 engineers on the consumer web platform and mobile app team (SEO, Core Web Vitals, conversion optimization, legal compliance), leading delivery on LCP regression fixes for high-traffic page types, INP tracking ahead of Google's CWV INP rollout, and an org-wide cookie consent rollout.
- Returned to the IC track in Jun 2023 to focus on data platform architecture.

LIFX / Buddy Technologies

(contractor through Bluehawk Consulting, converted to FTE in Jan 2020)

Software Engineer — Jul 2019 – Dec 2021

- Built a React Native app replacing separate native iOS and Android applications, and shipped features in the native apps to support a new LIFX power switch product.
- Worked with a team of three for four months to replace a fragile monolithic platform including a portal with a new microservice-based platform, saving \$120k a year while improving availability and maintainability.
- Built a React Native application for Airstream's Smart platform; led mobile development and authored custom native plugins for unique connectivity scenarios.
- Built tooling and test frameworks for code quality (type-checking, linting, unit testing) and regression/UX testing (Storybook, screenshot diffing, Detox E2E).
- Architected an end-to-end "checklist service" as a fully tested Micronaut microservice with CRUD support and a performance-optimized client API.
- *Technology used:* React Native, TypeScript, Java, Swift, Kotlin, Micronaut, PostgreSQL, Azure (AKS, DevOps), MQTT, HomeKit.

Belkin International

Senior Software Engineer — Nov 2018 – Jul 2019

- Extended a Cordova mobile application, built React Native and Flutter proof-of-concept apps, and created a test automation framework.

UpTop

Product Development Manager — Jun 2018 – Nov 2018

- Delivered a React Native application for TERRA Staffing and an Alexa skill for TD Bank.
- Laid the groundwork for a platform for the cleaning industry by building a two-person product-focused team.

Development Team Lead — Apr 2016 – Jun 2018

- Contributed as IC at 120% billable utilization while building a full-stack development team that grew to 12.

- Successfully delivered 13 web and hybrid mobile application projects for clients including Microsoft, Premera, Capital One, Belkin, and Entertainment Partners.
- Led architecture shift from server-rendered ASP.NET MVC to Web API + Angular on Microsoft's Cloud Launchpad portal, decoupling frontend and backend concerns.
- Served as lead mobile developer on the Belkin Linksys consumer app, with the team delivering on schedule for the CES launch window.
- *Technology used:* ASP.NET Core, C#, Node, MS SQL, Azure, AWS, React, Angular 2+, PHP, WordPress, Elasticsearch.

Software Developer — Jul 2015 – Apr 2016

- Finished three client projects (each estimated between 4 and 6 months) early and under budget.

Earlier Career (1997–2015)

Leadership roles spanning engineering, operations, sales and marketing, and general business management at manufacturing and industrial automation companies.

Innovations

Co-inventor on US patent 7,610,734 for siding installation tools, plus two related fiber-cement cutting applications, developed through 3D CAD, CNC prototyping, and 3D printing.

Education

Olympic College

Associates Degree (A.A.S.), General Studies — 3.75 GPA / Phi Theta Kappa